Validation of a Voice-Tone and Background-Noise Analyzer on Production Call Audio

Validation report · AutoAce AI technical trial

Abstract

A voice-tone and background-noise analyzer emits nine fields per call. This report asks which of them survive contact with real call audio, and every number below is measured on real recordings rather than on the lab-generated audio used during development. Four sources supply the evidence: 30 HarperValleyBank customer-service calls, 114 mixes of ESC-50 environmental noise into confirmed-clean speech at controlled signal-to-noise ratios, 150 IEMOCAP utterances used only for confusion analysis, and the three labeled clips provided with this trial. Two fields hold against real audio, one mostly holds, two drop, and four are unconfirmed. Background-noise detection holds at 96.5% accuracy and 0.82 F1, and noise-type identification mostly holds at 80.6% exact match. Emotional tone holds at 0.49 macro-F1, level with published prompt-based systems but short of production confidence. Speaker-overlap detection and noise-severity rating both drop, to 60% and 46.5% from 80% and 72.5% on synthetic benchmarks, and neither regression has been root-caused. The unconfirmed four include `audio_quality` and `long_silence_present`, for which no real-audio label was available, and `confidence`, which cannot be calibrated while all three provided clips share one value. The pipeline runs at $0.00036 per audio-minute, 12% of the $0.003 ceiling, and at 0.14–0.62× realtime single-threaded on CPU.

1. Evidence base

Every number in this report comes from one of the four sources in Table 1. Where a field has no real-audio check, it is marked *unconfirmed* rather than backed by a synthetic estimate.

Table 1: Real-audio evidence base.

Source	What it is	Fields it checks	Sample
HarperValleyBank	Real customer-service calls, US bank	`emotional_tone` (coarse), `speaker_overlap_present`	30 of 1,446 calls (seed 1234)
ESC-50 × clean speech	Real environmental noise mixed into confirmed-clean speech at controlled SNRs	`background_noise_present`, `_severity`, `_type`	114 mixes (type: 108)
IEMOCAP	Acted emotional speech, five classes	`emotional_tone`, `emotional_intensity` (confusion analysis only, not an accuracy claim)	150 utterances
Provided production calls	The three labeled clips from this trial	all fields	3 clips

2. Results by field

Table 2 reports, for every field in the output schema, the real-audio check (the production-relevant result), the development-time benchmark, and the verdict.

Table 2: Field-by-field results: real-audio check versus development benchmark. Verdicts: • holds (real audio confirms the field), • mostly holds, • drops (real audio contradicts it), • unconfirmed (no real-audio evidence either way).

Field	Real-audio check	Benchmark	Verdict
`emotional_tone` (pos/neu)	60% / 0.49 F1 real bank calls, n=30	0.49 macro-F1 IEMOCAP, n=150	• holds
`emotional_intensity`	no source (2/3 on the provided clips)	0.47 / 0.41 F1 IEMOCAP, n=150	• unconfirmed
`speaker_overlap_present`	60% / 0.58 F1 real calls, n=30	80% / 0.79 F1 synthetic	• drops
`background_noise_present`	96.5% / 0.82 F1 ESC-50 mixes, n=114	97.5% / 0.97 F1 synthetic	• holds
`background_noise_type`	80.6% exact match ESC-50, n=108	—	• mostly holds
`background_noise_severity`	46.5% / 0.51 F1 ESC-50 mixes, n=114	72.5% / 0.71 F1 synthetic	• drops
`audio_quality`	no real-audio label available	93.3% / 0.93 F1 synthetic	• unconfirmed
`long_silence_present`	no real-audio label available	95% / 0.95 F1 synthetic	• unconfirmed
`confidence`	uncalibratable (3 labels share one value)	—	• unconfirmed

2.1 Tone: where it gets confused

Thirty real calls cannot support a confusion matrix. At that sample size it is mostly noise. To see *how* tone predictions miss, the system was run on IEMOCAP, a public five-class acted-speech benchmark, purely to observe which emotion pairs get confused and how that compares with published research. This is not a claim about real-call accuracy. The real-call numbers are those in Table 2. Accuracy 0.493, macro-F1 0.494, n=150 (0.45 $\pm$ 0.03 macro-F1 across five seeds).

Table 3: Confusion matrix for `emotional_tone` on IEMOCAP (n=150).

	Predicted				
Actual	Neutral	Satisfied	Frustrated	Upset	Distressed
Neutral	**18**	1	9	2	0
Satisfied	9	**11**	5	3	2
Frustrated	4	2	**16**	7	1
Upset	0	1	7	**18**	4
Distressed	13	0	6	0	**11**

Rows are actual class. Columns are predicted class. Bold diagonal denotes correct predictions.

`frustrated` and `upset` are most often confused with each other and with `neutral`, the weakest boundary in the label set (per-class F1 0.44 and 0.60). Table 4 positions this system against the closest published result.

Table 4: Positioning against published IEMOCAP results (unweighted accuracy).

System	Transcript only	+ Context
Zero-shot baseline	43.4%	55.5%
SpeechCueLLM	50.0%	60.1%
VowelPrompt (published SOTA)	51.2%	62.3%
This system	**50.0%**	**55.7%** (54.7–56.7)

Two cautions bound that comparison. IEMOCAP's inter-annotator agreement is Fleiss' κ 0.27 across all classes and 0.48 on the four main ones, which is fair at best and caps every system evaluated against these labels. Our label set is also harder than the published one, because it separates `frustrated` from `upset` by degree of anger alone, the axis the literature identifies as hardest.

2.2 Noise: where it gets confused

One confusion is specific and explainable: `wind` is misclassified as road noise in 17 of 18 cases (94%). This is most likely a limitation of the upstream PANNs/AudioSet classifier rather than an error introduced by this pipeline. It is worth attention if wind matters to AutoAce's real call conditions.

The severity regression is less understood. Severity thresholds were calibrated on synthetic seed-1234 audio, which likely produces more acoustically uniform noise than real recordings provide. That calibration mismatch is the leading suspect for the 46.5% real-audio score, but it has not been root-caused.

2.3 The three provided calls

Tone matched on 1 of the 3 provided clips under every configuration tested, and `emotional_intensity` matched on 2 of 3. Three clips cannot confirm or refute anything, since each clip is worth 33 percentage points, but the result is reported because it is the only check on the exact audio this trial will be evaluated on.

3. Cost and latency

Table 5 reports cost and wall-clock processing, measured end to end on the three provided clips.

Table 5: Cost and latency on the provided clips.

Clip	Duration	Cost	$/audio-min	Processing	× realtime
call_001	30.9 s	$0.000464	$0.00090	19.2 s	0.62×
call_002	35.0 s	$0.000339	$0.00058	5.9 s	0.17×
call_003	171.9 s	$0.000616	$0.00022	23.4 s	0.14×

Weighted average cost is $0.00036 per audio-minute, 12% of the ceiling, on OpenAI `gpt-5.6-luna` at $0.20 / $1.20 per million input/output tokens and $0.02 per million cached input tokens. Only transcript text and numeric acoustic features are transmitted to that endpoint, and audio never leaves this infrastructure. Batch LLM calls run concurrently (1.64× measured wall-time speedup, n=8). Every other pipeline stage runs locally at no marginal cost.

4. Limitations and next steps

- `speaker_overlap_present` and `background_noise_severity` are the two weakest real-audio results and neither regression is root-caused. Next: sweep the overlap decision threshold on real calls, and recalibrate severity thresholds against the real ESC-50 mixtures rather than synthetic audio.
- `audio_quality` and `long_silence_present` have no real-audio check at all. The only candidate label for `audio_quality` (call intelligibility) measures a different construct, and no dataset on hand contains genuine dead-air calls. Next: source real labels, or redefine the fields against a label that can be obtained.
- `confidence` is an uncalibrated placeholder with nothing to calibrate against. Next: calibrate as soon as more than three labeled calls exist.
- Sample sizes are small: 3 clips, 30 calls, 114–150 mixes or utterances. These numbers describe this evaluation, not a statistically confident estimate of production performance.
- The `frustrated`/`upset` boundary is the axis this system and the published literature both find hardest, and is where tone accuracy is most improvable.

Data and code availability

Every result here is reproduced by the scripts in `scripts/` from the JSON artifacts in the repository root. Full methodology, covering validation design, ablations, seed replication, and the evidence behind every number, is in `TECHNICAL_MEMO.md` and `PLAN.md`.

References

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[5] Wu, M., Nafziger, J., Scodary, A., and Maas, A. HarperValleyBank: A domain-specific spoken dialog corpus. *arXiv:2010.13929*, 2020.